Recipe Agent

By: Jesigga Sigurdardottir, Sarosh Awan, Jalen Lee,
Jon Spikes, Pranav Vad, Randy Van Waes

Roles

Scrum Master - Jalen

Progress Reporting - Will be rotating per week

UI/Coding: Jalen, Pranav, Sarosh

Github Link:

https://github.com/saroshawan100/Recipe-Agent

Progress Update

What is New?

Image Generation

Output an image for the step when requested to save tokens

- Clears confusion by showing a target appearance
- Lets users compares to the food being made
- Adds visual clarity

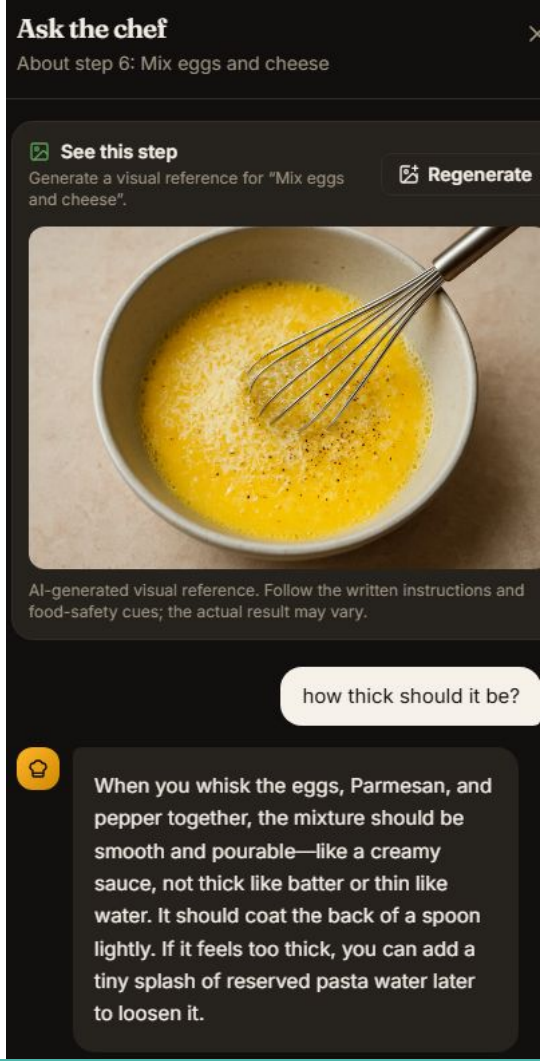
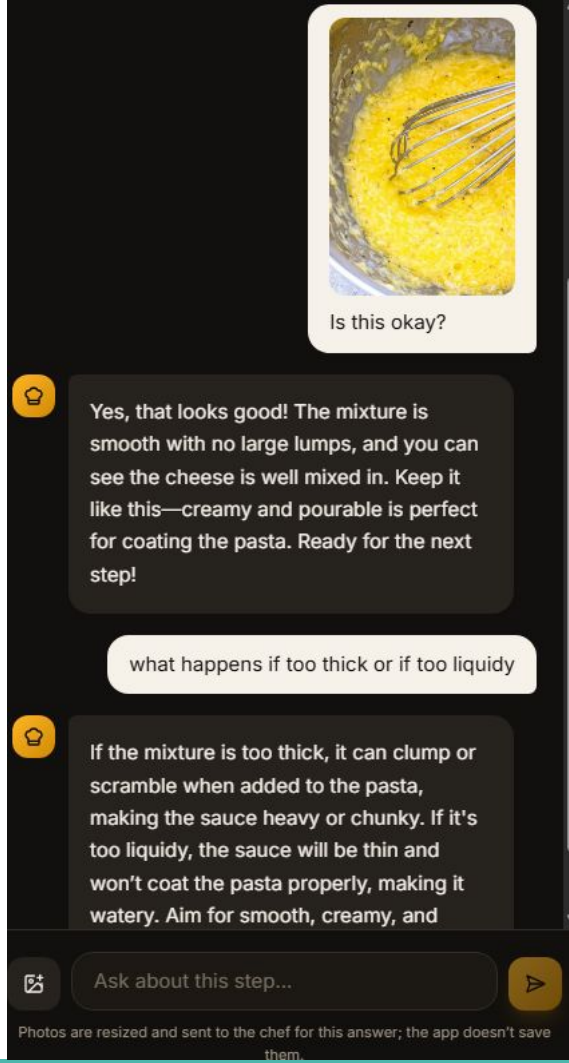


Image Recognition

Input images when users are lost and confused

- Recognizes what the food should look like in the current step
- Offers timely feedback to salvage and remedy the situation
- Compare visible cues to recipe’s instructions
- Help reassure users and provides confidence



Evaluation Framework & Key Findings

- Built an evaluation suite for the AI agent: 41 test cases across all six stages (recipe suggestions, full recipes, live coaching, check-ins, in-cook Q&A, chat). Each case is graded by automated rule checks for allergens, diets, time limits, safe cooking temperatures and units, plus an optional AI judge that scores answer quality against a rubric.

- Measured a baseline and found real safety issues: 29 of 41 cases passed (71%). The agent suggested peanut dishes to a user with a peanut allergy, used regular soy sauce in "gluten-free" recipes, ignored time limits on "quick shop" recipes, and left long gaps in live coaching.

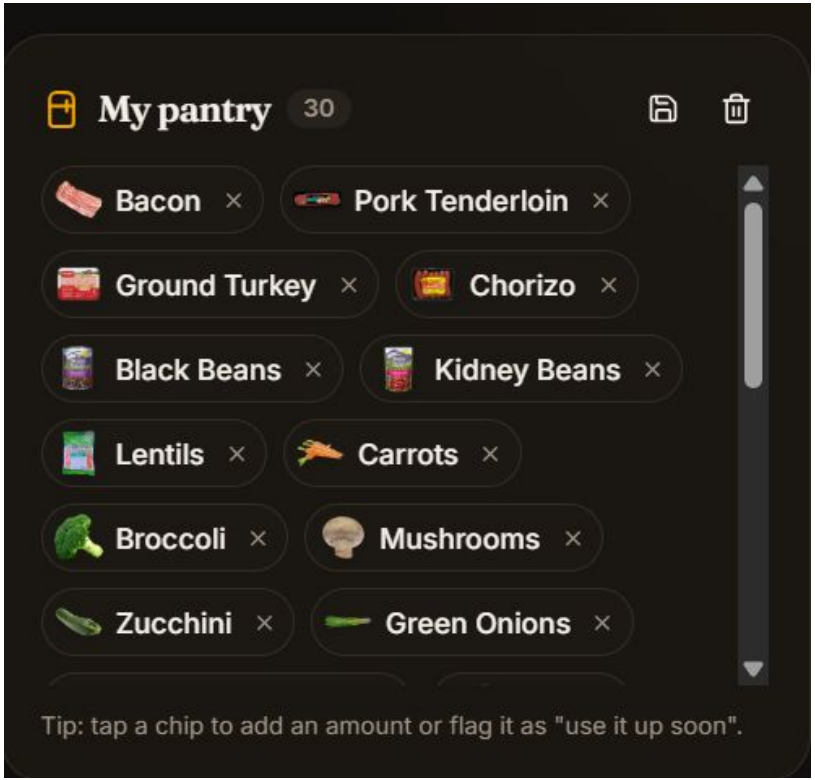
Accounts and Syncing

- **Personalized user accounts:** Added email and password signup and login, with secure password hashing and persistent sessions, so users can return to their own account.
- **Synced pantry and preferences:** Connected each account to its own pantry and cooking preferences, including allergies. The app restores that personal data after login and syncs changes as users update it.
- **MongoDB Atlas:** Added MongoDB storage for account, pantry, and preference data, configured through a private .env connection string. Verified that the backend connects to the Atlas cluster.

Accounts and Syncing Example



- Example of creating an account and preferences being saved to a specific user.



Proposed Next Steps

- **Fridge photo input:** let users upload a photo and detect ingredients automatically
- **Decide on the recipe bank:** keep it in scope, or rely on the Al model with structured outputs
- **Deploy:** host the app so the whole team and testers can use it without running it locally

Questions?

Thank you!